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Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738157390960-Chapter 9 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

WHAT YOU'LL DO IN THIS UNIT

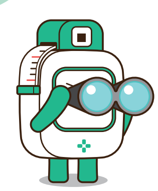
you'll take everything you learned about fire safety in Unit 1 and start creating your very own fire alarm system! This is where your ideas start to take shape. You'll brainstorm the best ways to detect fire and alert people, choose the right MODI modules for the job, and begin the process of designing and building your device.

IDEA GENERATION - THE COMPONENTS YOU'LL USE

1. Think About What Your Device Needs to Do

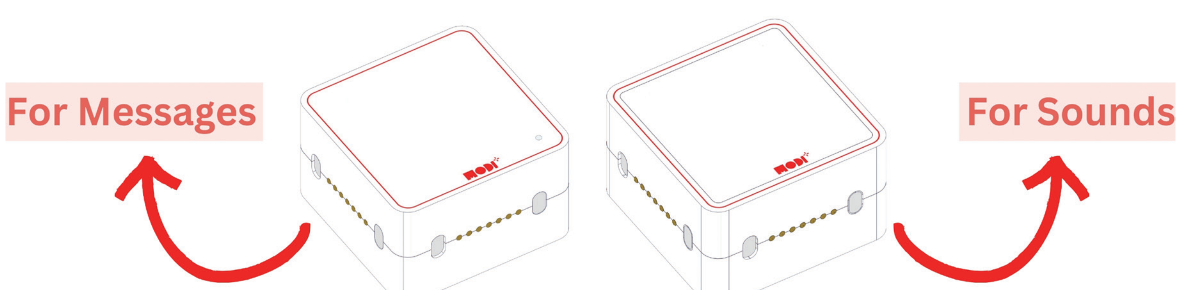
- Your fire alarm system needs to detect signs of a fire, like smoke or a sudden rise in temperature. How can you use technology to sense these signs? Think about the different ways fires can start and how your device can detect them quickly.

Since we can't detect smoke with the MODI modules, we need a sensor to detect how hot the space is. Thus, we need ________________ sensor!



2. Consider the Response

- Once your device detects a fire, it needs to alert people quickly. We want to use both visual and auditory signals to make sure everyone notices the alert. Think about how you could use a light or sound to grab attention and keep people safe..
- **Hint:** We are going to show a "Fire Detected" sign on a screen and play an "Alert" sound through a speaker.

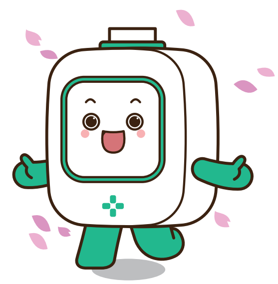


3. What Other Possible Components?

- Your fire alarm system will need a way to reset or turn off the alarm after it has detected a fire. Consider adding a component that allows someone to stop the alarm when the situation is safe again or if the system mistakenly detects a fire. How can this be done efficiently and safely?

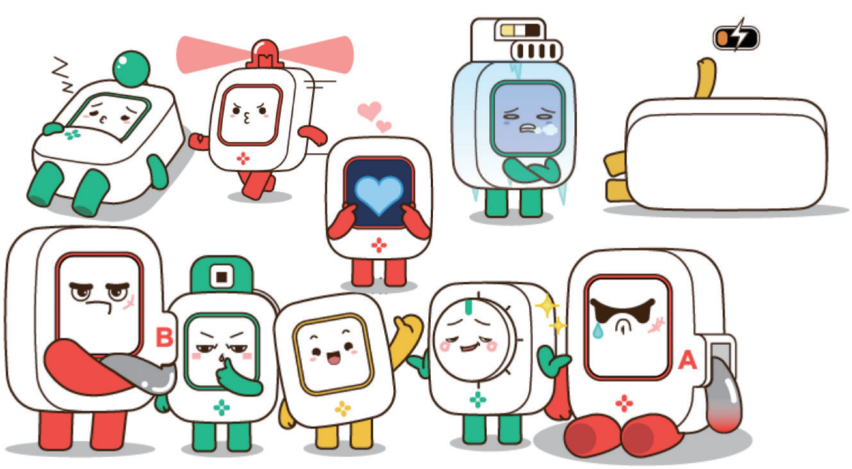
HINT

It will be most efficient if you can stop the alarm with just one touch.



Tip for Brainstorming

Think about the key parts of your Fire Alarm System:
a sensor to detect signs of fire, a way to process this information, and something to alert people. What technology could you use for these parts?

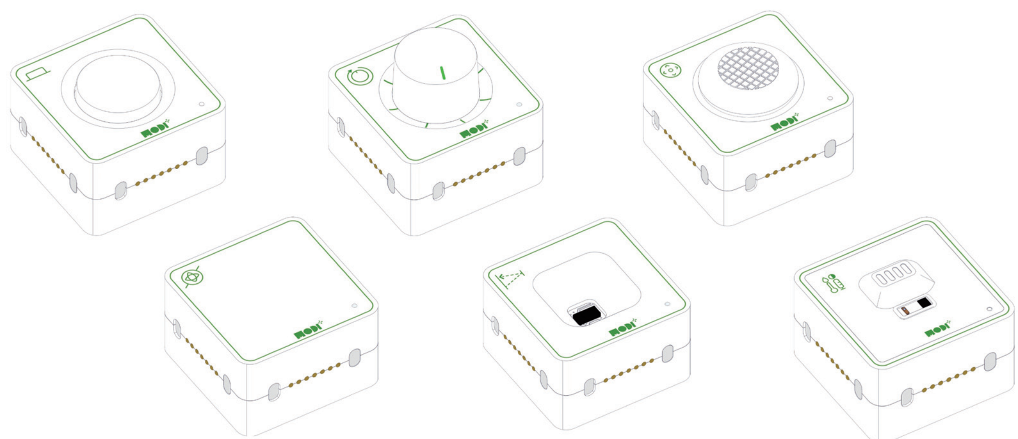


Required Modules & Brainstorming

To build our Flood Alarm System, we need to decide which MODI modules to use. Here are some ideas to get started:

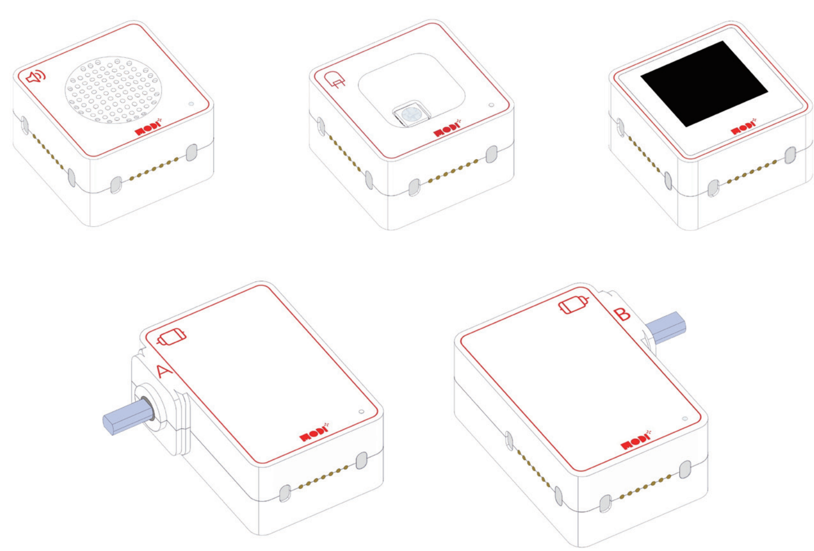
Sensor Modules

- Temperature Sensor: Think about which MODI module can detect the temperature in the room. This sensor will be crucial for identifying when the temperature rises above a safe level.



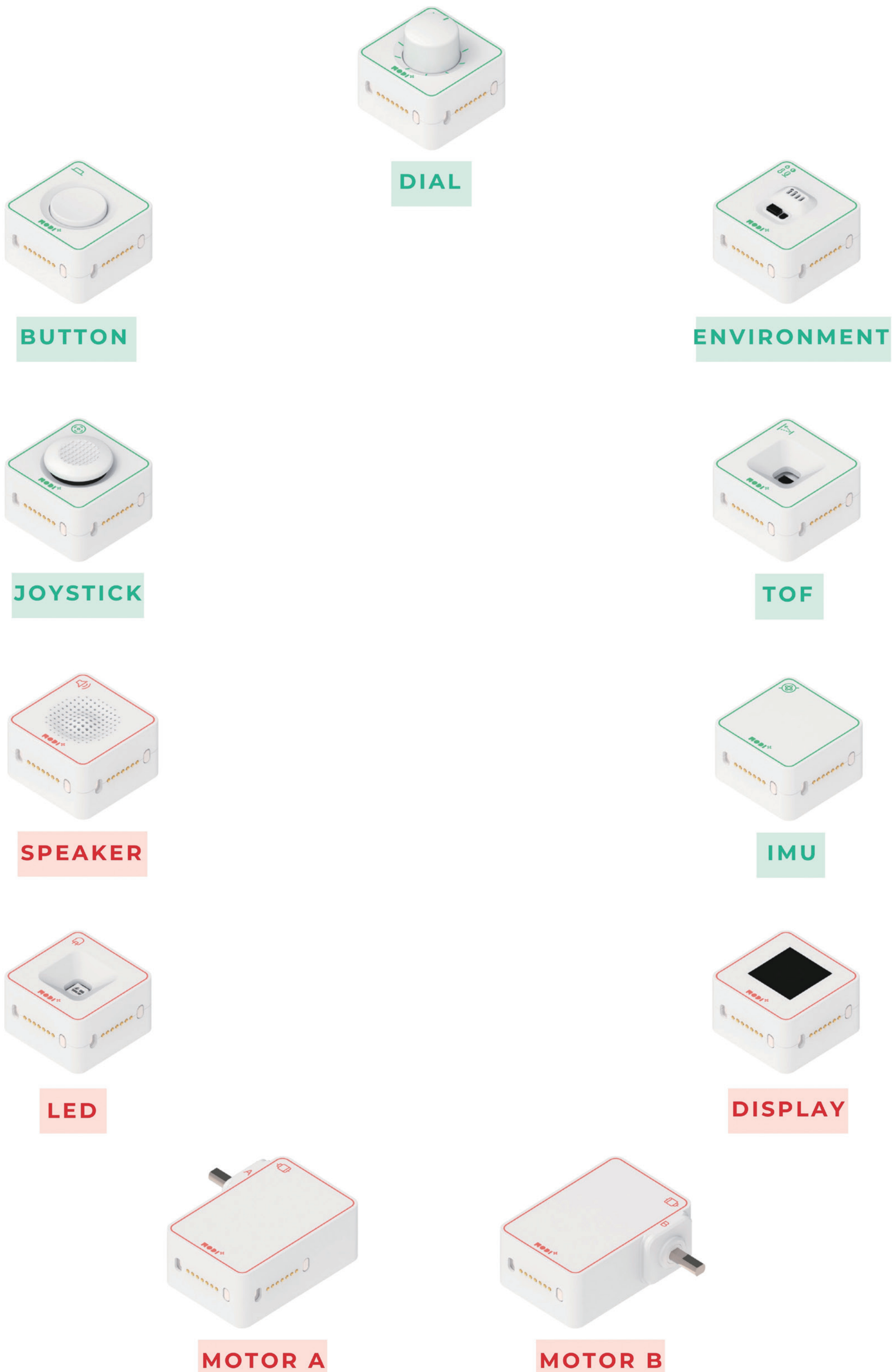
Output Modules

- Auditory Signal Module: Consider which module can emit sound to provide an auditory alert when a fire is detected.
- Display Module: This module can show important messages or status updates, such as indicating that a fire has been detected.



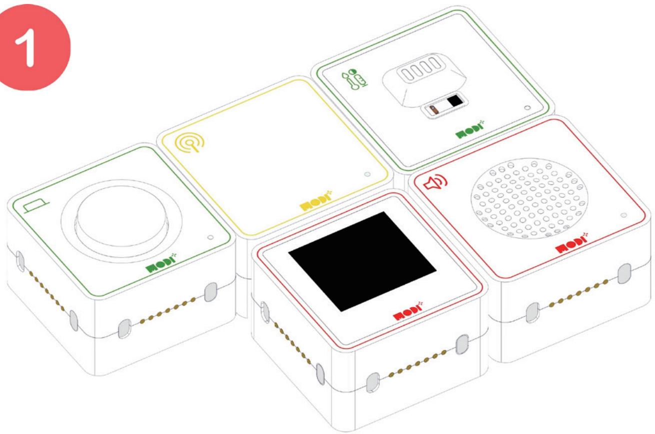
Required Modules & Brainstorming

Which MODI modules do you need for the creation?



WHEN WE COMBINE ALL THE MODULES, THE CREATION HAS TO BE LOOK LIKE THIS.

1



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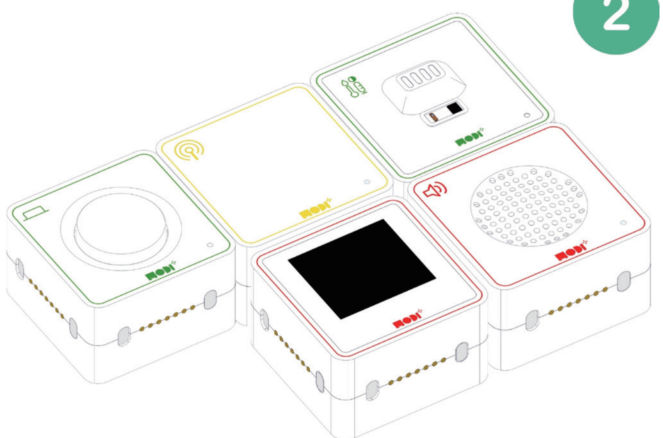

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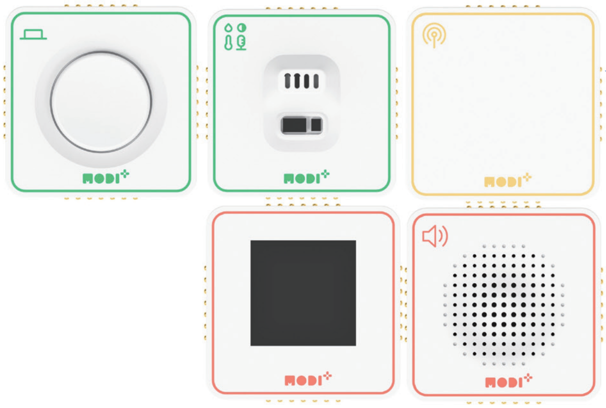
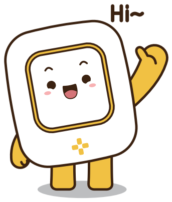


TIPS

If you have all the required modules, the physical layout doesn't matter much. Just make sure all the modules are properly connected!



Module Functionality

DISPLAY MODULE

How It Works: The Display Module shows important information through text or symbols. It can display real-time updates or alert messages depending on the situation.

Applications in Fire Alarm System

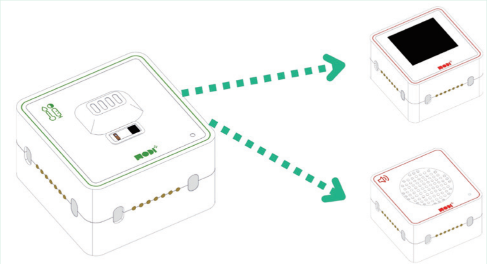
- Status Updates: The display shows the current temperature under normal conditions.
- Alert Messages: When the temperature exceeds 70°C, the display shows "Fire Detected" to warn people in the area.

ENVIRONMENT SENSOR

How It Works: The Environment Sensor monitors the temperature in the room. It constantly checks for any rise in temperature, especially above a certain threshold (e.g., 70°C), which could indicate a fire.

Applications in Fire Alarm System

- Fire Detection: When the temperature exceeds 70°C, the sensor triggers the alarm system to notify you of potential fire hazards.

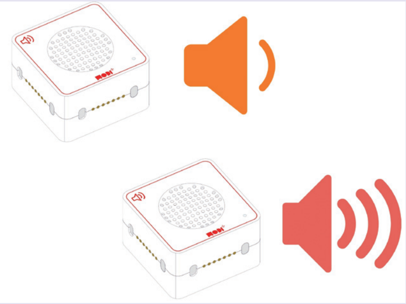


SPEAKER MODULE

How It Works: The Speaker Module emits sound alerts when triggered. It can produce different sounds to indicate various levels of urgency.

Applications in Flood Alarm System

- Warning Sound: When the temperature rises above the threshold, the speaker emits an alarm sound to alert everyone nearby.
- Continuous Alert: The speaker continues to sound the alarm until the situation is resolved or the button is pressed.

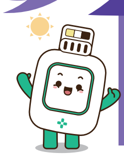


WHY THESE MODULES ARE EFFECTIVE IN A FIRE ALARM SYSTEM

The Environment Sensor, Display, Speaker, and Button modules are essential components of an effective fire alarm system. Each module plays a crucial role in ensuring the system is responsive, informative, and easy to use. Here's how they contribute:

Comprehensive Alerts

By combining visual (Display), auditory (Speaker), and interactive (Button) alerts, the fire alarm system effectively communicates in various situations and for different people. Some might see the warning message on the display first, while others might hear the alarm sound.

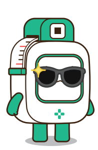


DISCUSSION POINT

Think about fire alarm systems you've seen or heard about. How do they alert people? What parts do they use to make sounds or show warnings?

Redundancy

Having multiple forms of alerts ensures that if one method is missed, others will still convey the necessary warnings. For example, if someone doesn't see the display message, they might hear the alarm or be able to reset the system with the button.

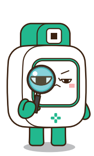


DISCUSSION POINT

Why is it important to have different ways to alert people in fire alarm systems? Can you think of an example where having both sound and visual alerts would be important?

Accessibility

The system is more accessible to people with different sensory abilities. For instance, visual alerts on the display help those who are hearing impaired, while sound alerts from the speaker assist those who might not be looking at the display. The button allows for easy interaction to stop the alarm, making the system user-friendly.



DISCUSSION POINT

Our society includes many different people, some with impaired sight or hearing. Think of one example where using different types of alerts can help people with sensory disabilities in an emergency.